
Billiard Balls

Consider an infinite flat frictionless surface. A yellow billiard ball of mass 4 kg is moving at a constant velocity on this surface. You're given N red billiard balls of mass 1 kg which you can place anywhere on the surface so that they collide with the yellow ball. Find the minimum N for the yellow billiard ball's velocity vector to make a 90° angle with its original velocity vector. All collisions are elastic and the tolerance is 0%.

The maximum deflection of a single collision is $\sin^{-1}(1/4)$ (prove it yourself! Hint: move into the center of mass frame. If you're stuck, consult problem 20(c) in Kevin Zhou's M3 handout). The answer is thus $\lceil \frac{90^\circ}{\sin^{-1}(1/4)} \rceil = 7$.